Skin Cancer Detection: From CNNs to Swin Transformers

A Comparative Architectural Analysis

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Abstract

The manual dermoscopic classification of skin lesions exhibits high inter-observer variability. This study evaluates an automated classification pipeline using the HAM10000 dataset, comparing baseline Convolutional Neural Networks (CNNs) against a Swin Transformer architecture. By integrating Shifted Window Self-Attention, a supplemental non-lesion control class, minority oversampling, and decoupled weight decay optimization (AdamW), the proposed Swin-T model demonstrates a validation accuracy of 92%. The results suggest computational viability for automated, multi-class dermatological analysis within constrained clinical latency parameters.

1 Introduction

The rising global incidence of skin malignancies necessitates efficient diagnostic methodologies. Standard dermoscopic analysis relies heavily on practitioner interpretation, introducing subjective variance and the potential for false-negative categorizations in rare malignancies. Consequently, there is a requirement for objective, high-sensitivity computational tools capable of processing clinical imaging within operational time constraints to support dermatological diagnostics.

2 Problem Formulation

The primary objective is the multi-class categorization of dermoscopic images utilizing the HAM10000 dataset. The operational framework necessitates addressing three primary constraints:

- Class Imbalance: The dataset exhibits significant skew; the melanocytic nevus (nv) class comprises ≈67% of the distribution, whereas minority malignant classes represent $< 2\%$.
- Diagnostic Sensitivity: The architecture must maintain sensitivity thresholds for malignant classifications (e.g., melanoma, basal cell carcinoma).
- Computational Efficiency: Inference latency is constrained to under 2.0 seconds to satisfy clinical deployment parameters.

3 Architectural Framework

The methodology integrates local feature extraction with global context modeling through the following mechanisms:

- Model Selection: Baseline architectures (ResNet-50, EfficientNet, YOLOv4, ViT) were evaluated against the Swin Transformer. The latter was selected for its $\mathcal{O}(M^2hwC)$ linear computational complexity derived from Shifted Window Self-Attention.
- Distribution Normalization: Dataset asymmetry was mitigated by incorporating a non-lesional control class (derived from ISIC 2020 data) and applying systematic minority oversampling.
- Optimization Strategy: A two-phase fine-tuning protocol (head stabilization followed by joint parameter adjustment) was implemented using the AdamW optimizer to apply isothermal L_2 regularization.

4 Empirical Results & Analysis

The empirical evaluation indicates measurable variances across the tested architectures. Convolutional baselines exhibited spatial limitation constraints, whereas standard ViT architectures were bounded by quadratic complexity limits. The Swin Transformer demonstrated superior synthesis of hierarchical local extraction with global attention.

Table 1: Architectural Performance Comparison

Architecture	Train Acc.	Val Acc.	Parameters
CNN (Baseline)	70%	62%	~2M
ResNet-50	85%	76%	25.6M
EfficientNet-B3	90%	82%	11.1M
YOLOv4	88%	79%	–
ViT-B/16	91%	83%	86M
Swin-T (Proposed)	97%	92%	29M

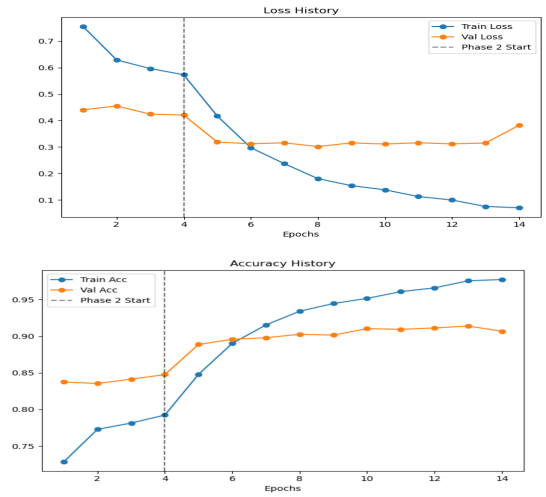

Figure 1: Training and Validation Loss (top) & Accuracy (bottom).

The final Swin-T configuration yielded a 92% validation accuracy. The integration of the control class effectively reduced false-positive rates on healthy tissue, while the decoupled optimization dynamics maintained required sensitivity parameters for critical malignancies.